BARCA INNOVATION HUB • METRICA SPORTS • TALENT ARENA

Zero-Phase Spearman Pitch Control & Kinematic Stoppage Engine

Automated Football Possession Tracking • More Than A Hack 2027 (Initial Challenge)

Team: LM10	**Fixture:** FC Barcelona vs Athletic Club (BAR-ATH, 199,319 frames)	**Pitch:** 105.3m × 68.0m (25.0 fps)	**Repo:** github.com/KevinNaserwan/football-possession-tracker

1. Problem Statement & The spellF1 Metric Challenge

The Initial Challenge requires classifying all **199,319 frames** of FC Barcelona vs. Athletic Club into four discrete states: `H` (Barcelona in possession), `A` (Athletic Club in possession), `D` (Dead Ball), or `X` (Abstention: pre-match, halftime gap, post-match).

The official ranking metric is **`spellF1`**, defined as the mean F1 score across five temporal Intersection-over-Union (IoU) matching thresholds ($\theta \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$). Standard per-frame distance heuristics produce catastrophic fragmentation: nearest-player algorithms predict 131 disjoint possession spells (vs. 17 true possessions), scoring a trivial `spellF1 = 0.0378` despite an apparent `macroF1 = 0.5972`. To win on `spellF1`, possession must be modeled as a continuous physical state with temporal inertia.

2. Methodology & Mathematical Architecture

2.1 Spearman Interception Surface (Δt)

For ball position r_b at frame t, each tracked player j with position r_j and velocity v_j requires arrival time t_j:

$$r_{j,react} = r_j + v_j \cdot \tau_{react} \quad | \quad d_j = \|r_{j,react} - r_b\|_2 \quad | \quad t_j = \tau_{react} + d_j / v_{max}$$

where τ_{react} = 0.5s (neuro-muscular reaction latency) and v_{max} = 7.5 m/s (maximum sprint speed). Team possession probability is governed by the logistic differential $P(\text{Home}) = 1 / (1 + \exp(-\lambda (t_A^* - t_H^*)))$ with sharpness $\lambda = 4.3$.

2.2 The 41% Missing Ball Trap & Possession Latch

The ball is absent (`NaN`) on **41.0% of graded gameplay frames** due to aerial clearances, penalty-box scuffles, and camera occlusions. While naive baselines default to `D` (causing severe possession splitting), our **Possession Latch** maintains continuous team possession across occlusions and passing trajectories unless a verified stoppage occurs.

2.3 Turnover Control Radius (≤ 2.2m) & Zero-Phase Backdating

Under competition rules, deflections without control do not transfer possession. We enforce:

- **Physical Reach Guard ($d_{opp} \le 2.2$m):** Opponents cannot initiate a turnover while the ball is traveling in the air or cleared downfield (d > 2.2m).
- **Confirmation Window (K = 12 frames, 0.48s):** Requires sustained control superiority before flipping state.
- **Zero-Phase Backdating:** Once confirmed, the transition timestamp is recorded at frame t - K + 1 (initial physical contact), cutting mean latency from +34 frames to **-2.3 frames (0.09s)** and achieving 100% transition recall (20/20 matched).

2.4 Multi-Modal Stoppage Segmentation (Dead Ball D)

Play is marked as `D` when: (1) ball crosses pitch boundaries (x < -0.01, x > 1.01, y < -0.01, y > 1.01), (2) a pre-restart set-piece interval is active (expanded 150 frames before restart to capture the preceding foul/whistle), or (3) collective player speed collapses (v_{mean} < 1.15 m/s, v_{max} < 2.80 m/s) outside active tactical phases.

3. Engineering Trade-Offs (Deliberately Dropped Assets)

Dropped Component	Initial Purpose	Technical Justification for Dropping	Engineering Benefit
Raw Video Pixel Decoding (.mp4)	YOLO / ByteTrack optical tracking	Decoding 199,319 1080p frames requires ~350 GB I/O and >6h compute. ATD stream already	Inference runtime dropped from 6 hours to **6.10 seconds** on CPU.

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		provides 25 fps centroids at 0.05m accuracy.	
Broadcast Stream (BAR-ATH_broadcast)	Multi-view perspective fusion	Broadcast video features pan/zoom cuts and partial player visibility (10–14 players). Tactical camera continuously covers all 22 players.	Eliminated camera-sync drift; preserved 100% planar coordinate consistency.
Deep Sequence Models (Bi-LSTM / Transformers)	Learned temporal state transitions	Public ground truth contains only 5 minutes of labels. Deep networks overfit this small slice and fail on the unobserved 85 minutes.	Physics-grounded model has **zero unconstrained weights**, ensuring robust full-match generalization.

4. Official Scoreboard & Ablation Study

Evaluated using official scorer `evaluation/run_evaluation.py` against official ground truth (7,500 graded frames, 17 possessions, 20 transitions):

Model / Submission	spellF1 (Rank)	macroF1	transF1 (τ=25)	Matched Trans	Mean Latency	Alive/Dead F1	Possessions	Mean Dur
baseline_raw (Published floor)	0.0378	0.5972	0.1410	11 / 20	-3.1f	0.8920	131	2.0s
baseline_smooth25	0.2868	0.6200	0.2034	6 / 20	+8.2f	0.8974	36	7.6s
baseline_smooth50 (Official floor)	0.3256	0.6207	0.0816	2 / 20	+4.0f	0.8990	26	10.8s
baseline_smooth125	0.4242	0.5812	0.1667	3 / 20	-7.7f	0.9012	16	18.5s
kevin_spearman_smooth50	0.5111	0.6790	0.3000	6 / 20	-10.3f	0.8851	19	13.1s
kevin_spearman_tuned_d	0.5744	0.8768	0.5652	13 / 20	-4.8f	0.9773	22	11.0s
LM10_v2_apex	0.7871	0.9800	0.7568	14 / 20	-3.3f	1.0000	14	17.2s
LM10 (Our Final Zero-Phase Engine)	**0.9714**	**0.9929**	**0.9756**	**20 / 20 (100%)**	**-2.3f**	**1.0000**	**18 (GT: 17)**	**13.4s (GT: 14.2s)**

Verified Achievement Highlights:

- **spellF1 = 0.9714:** +198% increase over official baseline floor (0.3256). Stable across all IoU thresholds (0.3 to 0.7: all 0.9714).
- **100% Transition Recall:** Matched 20 of 20 transitions with only -2.3 frames (0.09s) latency via zero-phase backdating.
- **Tactical Tempo Harmony:** Predicts 18 possessions of mean duration 13.4s, aligning with real ground truth (17 possessions, 14.2s).

5. Reproduction & Execution

End-to-end execution generates `LM10.csv` in **6.10 seconds**:

```
git clone https://github.com/KevinNaserwan/football-possession-tracker.git
cd football-possession-tracker
python run_solver.py
python evaluation/validate_submission.py LM10.csv
```